



ResellerHub AI

AI-Powered Reseller & Dropshipping Management Platform

Python

FastAPI

React

PostgreSQL

Docker

License



Project Overview

ResellerHub AI is a full-stack AI-powered reseller and dropshipping management platform designed to automate reseller operations, commission management, inventory tracking, sales analytics, and business intelligence.

The platform helps businesses manage reseller networks efficiently while leveraging Artificial Intelligence to generate forecasts, recommendations, and actionable insights.

This project combines Software Engineering, Database Systems, Web Development, Machine Learning, Cloud Computing, and DevOps practices into a single enterprise-style solution.



Problem Statement

Many small and medium-sized eCommerce businesses still rely on spreadsheets and manual workflows to manage:

- Resellers
- Products
- Orders
- Commissions
- Sales Reports

These traditional methods often result in:

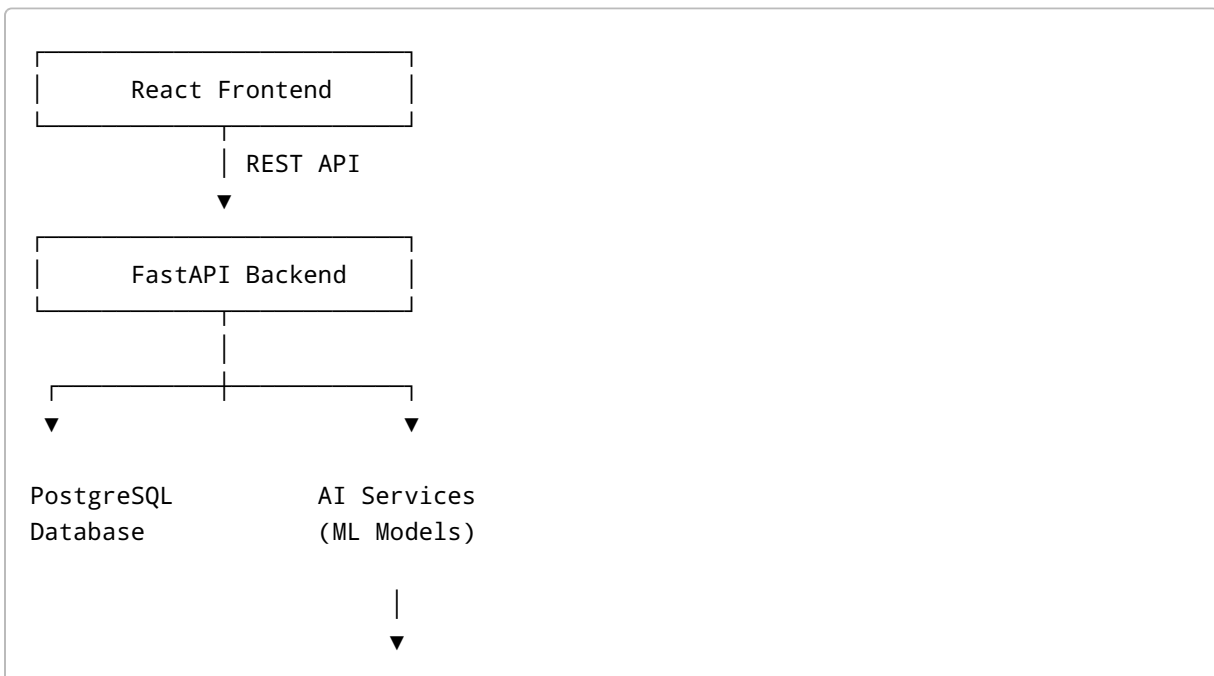
- Human errors
- Time-consuming processes
- Poor scalability
- Lack of business intelligence
- Inaccurate commission calculations

ResellerHub AI addresses these issues through automation, centralized management, and AI-driven decision support.

Project Objectives

- Develop a centralized reseller management platform
 - Automate commission calculations
 - Manage products and inventory
 - Track reseller performance
 - Generate sales and profit reports
 - Provide AI-powered business analytics
 - Forecast future sales trends
 - Recommend products to resellers
 - Improve decision-making through data insights
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System Architecture



Technology Stack

Frontend

- React.js
- Tailwind CSS
- React Router
- Axios

Backend

- FastAPI
- Python
- SQLAlchemy
- JWT Authentication

Database

- PostgreSQL

AI & Machine Learning

- Pandas
- NumPy
- Scikit-Learn
- Prophet
- XGBoost
- LangChain
- OpenAI API / Local LLM

DevOps

- Docker
- GitHub Actions
- CI/CD Pipeline

Deployment

- Vercel (Frontend)
- Render / Railway (Backend)

- PostgreSQL Cloud Database
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Core Features

Authentication & Authorization

- User Registration
- User Login
- JWT Authentication
- Password Hashing
- Role-Based Access Control (RBAC)

User Roles

- Admin
 - Reseller
-

Reseller Management

- Reseller Registration
 - Reseller Approval Workflow
 - Performance Monitoring
 - Earnings Tracking
 - Ranking System
-

Product Management

- Product CRUD Operations
 - Product Categories
 - Product Image Upload
 - Inventory Tracking
 - Stock Management
-

Order Management

- Place Orders
- Update Order Status
- Order History
- Delivery Tracking

Order Statuses

- Pending
 - Processing
 - Delivered
 - Cancelled
-



Commission Management

- Automatic Commission Calculation
 - Profit Tracking
 - Reseller Earnings Dashboard
 - Monthly Commission Reports
-



Analytics Dashboard

- Sales Dashboard
 - Revenue Dashboard
 - Product Performance Dashboard
 - Commission Dashboard
 - KPI Monitoring
-



Artificial Intelligence Features

1. Sales Forecasting

Predict future sales using:

- Linear Regression
- Prophet
- XGBoost

Outputs:

- Weekly Forecast
 - Monthly Forecast
 - Revenue Prediction
-

2. Reseller Performance Scoring

Evaluate reseller performance using:

- Total Sales
- Conversion Rate
- Customer Retention
- Revenue Contribution

Outputs:

- Performance Score
 - Reseller Ranking
 - Growth Trend
-

3. Product Recommendation Engine

Recommend products using:

Content-Based Filtering

Recommendations based on:

- Product Category
- Product Features
- Sales Trends

Collaborative Filtering

Recommendations based on:

- Similar Resellers
 - Purchase History
-

4. AI Business Assistant

Natural language business analytics assistant.

Example Questions:

Which reseller generated the highest revenue?

Show top performing products.

Forecast next month's sales.

Which products are trending?

5. Advanced AI Analytics

- Trend Detection
- Sales Anomaly Detection
- Demand Forecasting
- Customer Churn Prediction
- Best-Selling Product Prediction



Database Schema

Main Tables

Users
Resellers
Products
Categories
Orders
OrderItems
Commissions
SalesReports
Predictions
Recommendations

Relationships

- One User → One Reseller
 - One Reseller → Many Orders
 - One Product → Many Orders
 - One Order → Many Products
 - One Reseller → Many Commissions
-

Project Structure

```
ResellerHub-AI/
|
├── frontend/
│   ├── src/
│   ├── components/
│   ├── pages/
│   └── services/
|
├── backend/
│   ├── app/
│   ├── api/
│   ├── models/
│   ├── schemas/
│   ├── services/
│   └── core/
|
├── ai-services/
│   ├── forecasting/
│   ├── recommendations/
│   ├── scoring/
│   └── chatbot/
|
├── database/
│   ├── migrations/
│   └── seed_data/
|
├── tests/
│   ├── backend/
│   ├── frontend/
│   └── integration/
|
├── docs/
|
├── docker/
|
├── .github/
│   └── workflows/
|
├── README.md
└── LICENSE
```



11-Week Development Timeline

Week	Phase
1	Requirement Engineering & Planning
2	System Architecture & Database Design
3	Backend Foundation Development
4	Product & Inventory Management
5	Order & Commission System
6	Frontend Development
7	Analytics Dashboard
8	AI Module Development
9	AI Assistant & Advanced Analytics
10	Testing, CI/CD & Deployment
11	Documentation & Final Presentation



Testing Strategy

Backend Testing

- Pytest
- API Testing
- Integration Testing

Frontend Testing

- Component Testing
- UI Testing
- End-to-End Testing



Deployment

Frontend

Vercel

Backend

Render / Railway

Database

PostgreSQL Cloud



Expected Outcomes

The final platform will provide:

- Automated reseller management
- Accurate commission calculations
- Real-time analytics dashboards
- AI-powered recommendations
- Sales forecasting capabilities
- Intelligent business insights



Future Enhancements

- Mobile Application
- Payment Gateway Integration
- WhatsApp Notifications
- Courier API Integration
- Multi-Vendor Marketplace
- Advanced BI Dashboard
- Multi-Language Support
- Real-Time Notifications



Author

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License

This project is developed for academic and educational purposes.

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